Telling Stylized Avatars Apart at a Glance: Contour Height Maps and Silhouette Overlap for Hairstyle Design in Shared XR

Y.T.
Independent researcher

Abstract

In shared extended reality (XR), people meet many avatars at once and often see them small, far away or from behind. Stylized "cute" faces make this harder: the cues that make a face read as cute (a round outline, a large forehead, a small nose and chin) also make faces resemble one another. We describe a small, reproducible design method that separates the two jobs. Faces are modelled as height maps and studied through their contour lines, built from a reference face step by step (outline, masses, nose and brow, mouth and chin, hairline). Distinguishability is measured as the intersection-over-union (IoU) of binary silhouettes. For four stylized figures, faces alone overlap with IoU 0.87–0.95; giving each figure one dominant hairstyle silhouette (ponytail, bob, two buns, cloud) lowers every pair to 0.62–0.73. Texture-like styles such as braids barely change the silhouette (IoU 0.87 against no hair) and work as a second layer, not as the dominant shape. All numbers and figures are produced by an open script from the same code as an interactive tool.

1 Introduction

Shared, social XR—the "eXtended meta/uni/Verse" (XV) in the terminology of Mann et al. [5]—puts many avatars in one view. Recognising who is who at a glance matters there in a way it does not for a single character on a screen. Character designers often check this with a "squint test": shrink the character until only its filled silhouette is left, and see whether it can still be told apart from the others.

This note turns that habit into a number and applies it to a specific tension. Stylized avatars are frequently designed to look cute. The baby-schema cues associated with cuteness [4, 1]—a rounder face, a larger forehead, a small nose and chin—push faces toward one shared shape. We show, on a deliberately simple model, how much this costs in distinguishability and how a single dominant hairstyle per figure buys it back.

Our contributions are modest and practical: (i) a height-map face model whose personality is added coarse to fine, so each step can be inspected as a change in contour lines; (ii) a silhouette-overlap measure and a working threshold for "too similar"; (iii) measurements for four stylized figures and eight hairstyle types; and (iv) an open, reproducible script.

2 Method

2.1 Faces as height maps

A face is a height map $z(x, y)$: a dome of half-width r_x and half-height r_y, $z_0 = 0.9\sqrt{1 - (x/r_x)^2 - (y/r_y)^2}$, plus Gaussian bumps for forehead, cheeks, jaw, brow, eye sockets, nose, mouth and chin, and a plateau above the hairline. The heights are a drawing model, not anatomy. Personality is added in five steps, from wide contour spacing to narrow: (1) outline (r_x, r_y); (2) masses (forehead, cheeks, jaw); (3) nose and brow; (4) mouth and chin; (5) hairline. At step s a face uses

its own parameters for steps $\leq s$ and the reference face's for the rest, so the difference map $\Delta z = z_{\text{face}} - z_{\text{ref}}$ shows each layer arriving. Contour lines are extracted with marching squares, the two-dimensional case of marching cubes [3]; the two saddle cases are resolved by the cell's centre value.

2.2 Hair as height on the same map

Hair is added to the same height map rather than placed as a separate object: a tower above the crown, horizontal wings at temple height, a bob as a helmet-shaped plateau with a fringe, braids as narrow ridges, a ponytail as a tapered tube from the crown past the shoulder, two ring-shaped buns, and a cloud whose rim wobbles with two sine terms. A detail layer adds the parts that vary most between styles: the fringe type (blunt, swept, covering one eye, see-through), the parting ratio, and the lengths of the left and right side locks. The head is sampled on a 90×110 grid over $x \in [-1.35, 1.35]$, $y \in [-1.3, 1.75]$.

2.3 Silhouette overlap

A silhouette is the set of grid cells with $z > 0.02$. For two silhouettes A and B aligned on the same grid, distinguishability is measured by the Jaccard index [2], $\text{IoU}(A, B) = |A \cap B|/|A \cup B|$; 1 means identical. We use $\text{IoU} \geq 0.75$ as a working flag for "too similar". This threshold is a design heuristic chosen from inspecting the small filled silhouettes, not a perceptual result.

3 Results

3.1 Cute faces converge

Table 1 lists the outlines of the four figures; each keeps one outline cue of its own (round, roundest and shortest, slim oval, heart-shaped). Still, with faces alone every pair overlaps strongly, with IoU between 0.87 and 0.95 (Table 2, grey bars in Fig. 1, top row of Fig. 3).

Figure	r_x	r_y	r_y/r_x
Ponytail	0.84	0.98	1.17
Bob	0.83	0.94	1.13
Buns	0.74	1.04	1.41
Cloud	0.80	1.00	1.25

Table 1: Face outlines (step 1). Figures are named by their dominant hairstyle.

3.2 One dominant hairstyle per figure

Giving each figure a different dominant hairstyle (ponytail, bob, buns, cloud) plus its detail layer lowers every pair below the flag, to 0.62–0.73 (Table 2, black bars in Fig. 1; contours in Fig. 2). An earlier assignment in which two figures both had round masses (bob and cloud) left that pair at 0.81; separating them by direction—a flat, low helmet against a cloud raised above the crown—resolved it. In practice, dominant shapes are easiest to separate when they differ in direction: round, tall, wide, or off to one side.

3.3 Dominant shapes and textures

Fig. 4 isolates the hair: every style on the same reference face. Braids overlap a bare head at 0.87 and wings at 0.87: they change the surface, not the outline, and are better used as a

Pair		Face only	With hair
Ponytail	Bob	0.95	0.70
Ponytail	Buns	0.88	0.68
Ponytail	Cloud	0.95	0.66
Bob	Buns	0.87	0.62
Bob	Cloud	0.94	0.65
Buns	Cloud	0.92	0.73

Table 2: Silhouette IoU for every pair of the four figures.

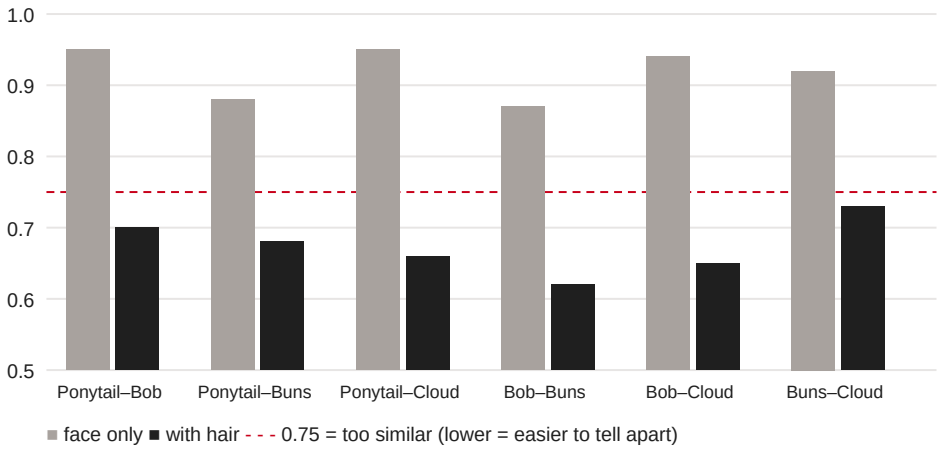


Figure 1: IoU per pair: faces only (grey) and with hair (black). The dashed line marks the working threshold 0.75.

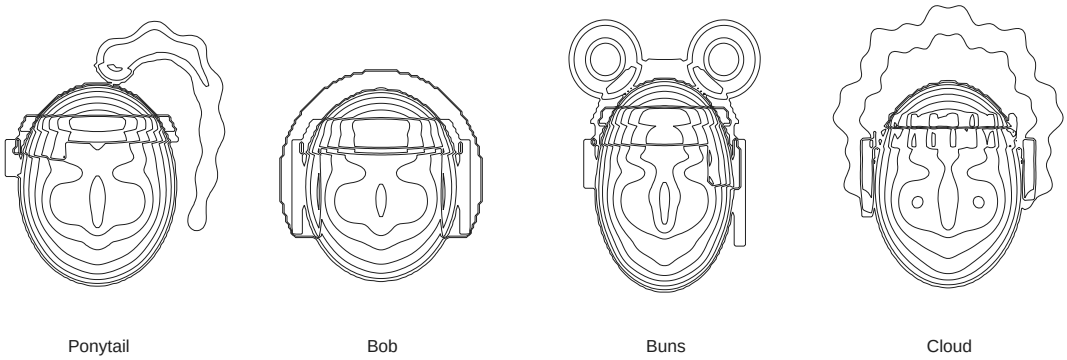


Figure 2: Contour lines of the four figures with hair (ten levels each).

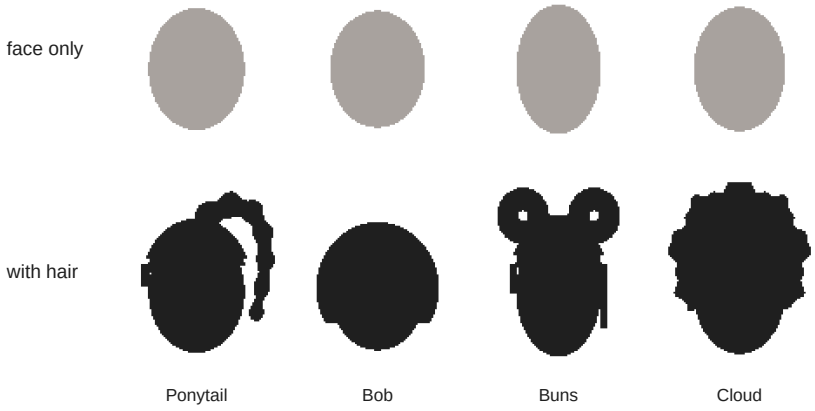


Figure 3: The squint test: filled silhouettes, faces only (top) and with hair (bottom).

second, texture layer. The cloud is the most distinct from a bare head (0.52). Variants of one style behave the same way (Table 3): a blunt cut that widens the outline (0.76 against the plain bob) or an asymmetric cut (0.84) changes the silhouette, while an inward curl or layering (0.92) mostly does not.

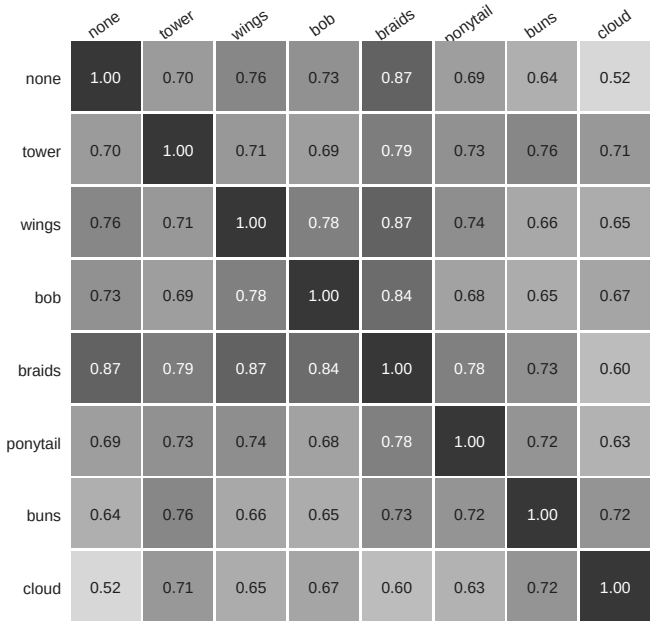


Figure 4: IoU between hairstyles on the same reference face (darker = more alike).

4 Discussion and limitations

The measure is deliberately crude. Silhouettes are taken from the front only, while avatars in XR are seen from every side; the next step is to average IoU over many viewpoints (for example, 22 cameras on a ring at three heights). The face model is a sum of smooth bumps, not anatomy. The 0.75 threshold has not been validated with people; a small recognition study (match a

Bob variant	IoU vs. plain bob
`bob-asym`	0.84
`bob-inward`	0.92
`bob-blunt`	0.76
`bob-layer`	0.92

Table 3: Variants of one style: which ones change the silhouette.

silhouette to a figure, under time pressure) would turn it into a perceptual result. Colour, which also separates avatars, is deliberately left out so that shape alone is tested.

5 Reproducibility

All numbers, tables and figures in this note are generated by `scripts/paper-figures.mjs` from the same code (`lib/faceContour.ts`) as an interactive tool in which the steps, hairstyles, fringe ratios and side locks can be changed and the IoU table updates live.

References

[1] Melanie L. Glocker, Daniel D. Langleben, Kosha Ruparel, James W. Loughead, Ruben C. Gur, and Norbert Sachser. Baby schema in infant faces induces cuteness perception and motivation for caretaking in adults. *Ethology*, 115(3):257–263, 2009.

[2] Paul Jaccard. The distribution of the flora in the alpine zone. *New Phytologist*, 11(2):37–50, 1912.

[3] William E. Lorensen and Harvey E. Cline. Marching cubes: A high resolution 3D surface construction algorithm. *ACM SIGGRAPH Computer Graphics*, 21(4):163–169, 1987.

[4] Konrad Lorenz. Die angeborenen Formen möglicher Erfahrung. *Zeitschrift für Tierpsychologie*, 5(2):235–409, 1943.

[5] Steve Mann et al. Beyond the Metaverse: XV (eXtended meta/uni/Verse). arXiv:2212.07960, 2022.